

Module 3

Sinusoidal Steady-State Analysis

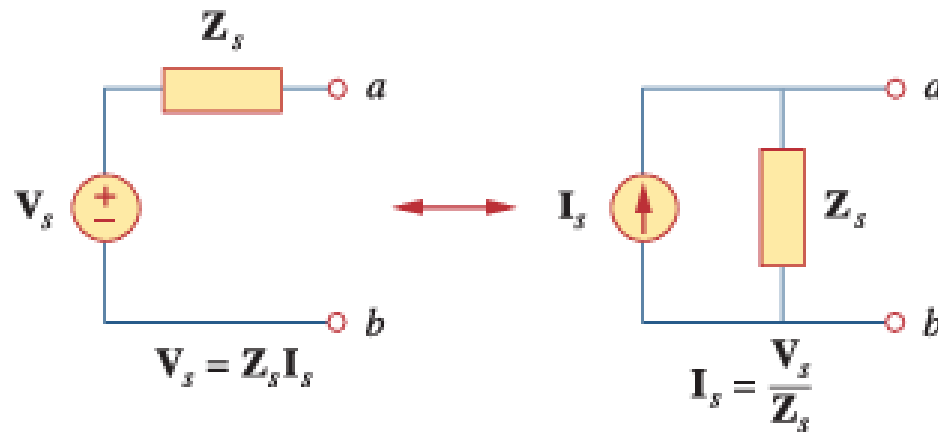
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Source Transformation

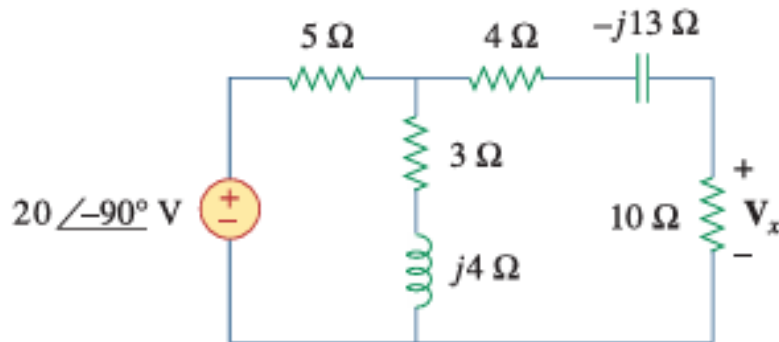
Source transformation in frequency domain involves transforming a voltage source in series with an impedance to a current source in parallel with an impedance or vice versa.

$$\mathbf{V}_s = \mathbf{Z}_s \mathbf{I}_s \quad \Leftrightarrow \quad \mathbf{I}_s = \frac{\mathbf{V}_s}{\mathbf{Z}_s}$$



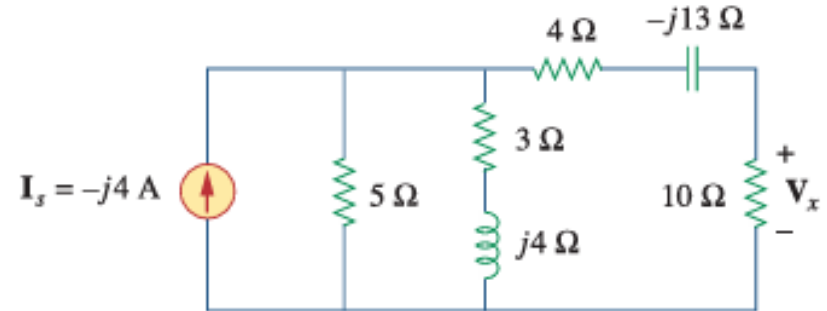
Source Transformation

1. Calculate V_x in the circuit using the method of source transformation.

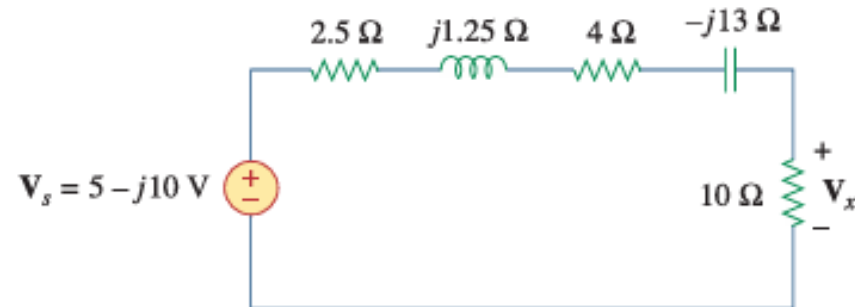


$$\mathbf{Z}_1 = \frac{5(3 + j4)}{8 + j4} = 2.5 + j1.25 \, \Omega$$

$$\mathbf{I}_s = \frac{20 \angle -90^\circ}{5} = 4 \angle -90^\circ = -j4 \, \text{A}$$



$$\mathbf{V}_s = \mathbf{I}_s \mathbf{Z}_1 = -j4(2.5 + j1.25) = 5 - j10 \, \text{V}$$

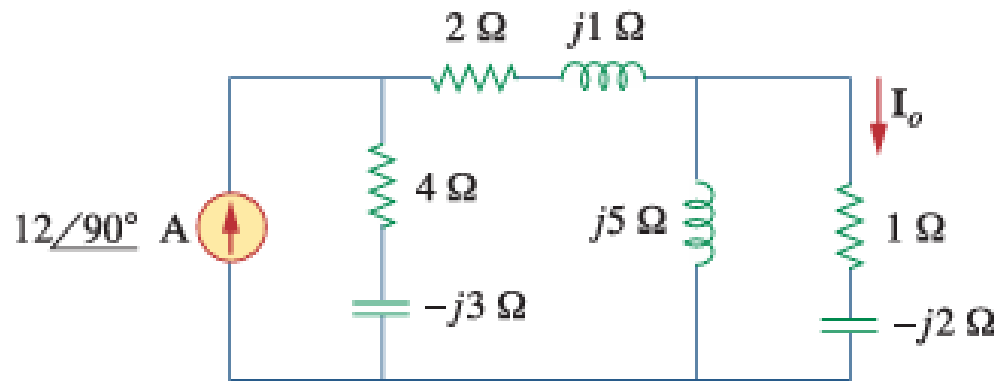


By voltage division,

$$\mathbf{V}_x = \frac{10}{10 + 2.5 + j1.25 + 4 - j13} (5 - j10) = 5.519 \angle -28^\circ \, \text{V}$$

Source Transformation

2. Find I_o in the circuit using the concept of source transformation.



Answer: 9.896 L99.46

Next class: Thevenin's and Norton's Theorem